SHOBHIT UNIVERSITY

Shobhit Institute of Engineering & Technology (Deemed-to-be University)

School of Biomedical Engineering & Health Sciences

NH-58, Modipuram, Meerut – 250110, Uttar Pradesh, India

Student Preparation Guide

From Cells to Circuits | Biomedical Engineering Workshop | July 7–11, 2026

Workshop: **7–11 July 2026** Daily: **10:00 AM – 1:00 PM** Venue: **Shobhit University Campus, Meerut** **Class XII Science Students**

Dear Participant, Welcome to the **From Cells to Circuits** workshop! This guide will help you prepare for 5 days of exciting hands-on learning. You do **NOT** need any prior knowledge — everything will be taught from scratch. However, reviewing these basics will make your experience richer and more enjoyable.

What to Review — Day by Day

Day 1 Prep: Cell Biology & Microscopy

Estimated prep time: 30–40 minutes

- **Cell structure:** Know the difference between animal and plant cells. Review the functions of nucleus, mitochondria, cell membrane, and cytoplasm.
- **Microscopy basics:** Understand what a compound microscope is. Know the terms: objective lens, eyepiece, magnification, focal length.
- **Blood composition:** Review what blood is made of — plasma, RBC, WBC, platelets. Know what blood groups are (ABO system).
- **Antisera & agglutination:** Understand why mixing wrong blood groups causes clumping (agglutination). This is the basis of the blood grouping experiment.

HELPFUL SEARCH TOPICS:

Cell organelles class 11 | How to use a microscope | ABO blood group system | What is agglutination

Day 2 Prep: Biosignals & ECG

Estimated prep time: 30–40 minutes

- **The heart's electrical system:** The heart generates electrical pulses with each beat. The sinoatrial (SA) node is the natural pacemaker. This signal travels through the AV node and Purkinje fibres.
- **What is an ECG?** An electrocardiogram (ECG/EKG) records the electrical activity of the heart over time, captured using electrodes placed on the skin.
- **P-QRS-T wave:** Each heartbeat produces a specific waveform. The P wave = atrial contraction; QRS complex = ventricular contraction; T wave = ventricular relaxation.
- **Electrodes & leads:** Standard 12-lead ECG placement — understand that leads are different viewpoints of the same electrical signal, not separate sensors.

HELPFUL SEARCH TOPICS:

How does ECG work | ECG P QRS T wave explained | Cardiac electrical conduction system

⚡ Day 3 Prep: Electronics & Circuits

Estimated prep time: 30–40 minutes

- **Basic electronics:** Understand Ohm's Law (V = IR). Know what resistors, capacitors, and op-amps do in a circuit. Understand voltage and current.
- **Breadboard:** A breadboard is a reusable circuit assembly tool. Understand the layout — horizontal power rails and vertical component rows.
- **Amplification:** Biological signals are very weak (microvolts). An amplifier increases signal strength. An instrumentation amplifier has high input impedance and rejects noise.
- **Signal noise:** EMI (electromagnetic interference) can corrupt biosignals. Filters remove unwanted frequencies. Know low-pass vs high-pass filters conceptually.

HELPFUL SEARCH TOPICS:

Ohm's Law explained | What is an op-amp amplifier | How to use a breadboard | Instrumentation amplifier basics

💻 Day 4 Prep: Arduino & MATLAB

Estimated prep time: 30–40 minutes

- **What is Arduino?** A microcontroller board that can read sensor data and control outputs. It uses a simplified version of C++. No prior coding experience needed.
- **Analog vs Digital signals:** Biological signals are analog (continuous). Arduino reads analog values (0–1023) via its ADC pins. Digital = on/off only.
- **What is MATLAB?** A mathematical computing software. We'll use it to plot ECG data and apply digital filters. Think of it as a powerful scientific calculator with graphs.
- **Data visualization:** How graphs help doctors — a waveform on a computer screen replaces a paper strip. Real-time plotting of biosignals is the foundation of patient monitoring.

HELPFUL SEARCH TOPICS:

Arduino for beginners | What is MATLAB used for | Analog vs digital signals

🤖 Day 5 Prep: AI in Healthcare

Estimated prep time: 20–30 minutes

- **AI & Machine Learning basics:** Understand that AI learns patterns from data. In medicine, AI can detect diseases from images, sounds, and signals faster than humans in some cases.
- **AI in ECG analysis:** Modern AI can detect arrhythmias, heart attacks, and other cardiac conditions from ECG waveforms. These tools assist (not replace) doctors.
- **Think of a problem:** For the Day 5 Innovation Challenge, think about a real-world health problem you'd like to solve using technology. Prepare a brief idea to discuss with your team.

HELPFUL SEARCH TOPICS:

AI in healthcare examples | Machine learning for beginners | Wearable health devices

General Tips for the Workshop

✅ **Do:** Come with an open mind. Ask questions — faculty encourage curiosity. Take notes. Work in teams and help classmates. Document your lab observations.

⚠️ **Lab Safety Rules:** No food or drinks inside the lab. Always wear your lab coat. Handle glass slides and sharp instruments with care. Report any spillage immediately to the instructor. Do not touch electrical circuits while the power is on.

Pre-Workshop Checklist

- [ ] Registration form submitted
- [ ] College ID card / Aadhar copy ready
- [ ] Lab coat arranged (or request at venue)
- [ ] Notebook & pen packed

- [ ] Read Day 1 & Day 2 prep topics
- [ ] Comfortable closed-toe footwear
- [ ] Water bottle
- [ ] WhatsApp group joined for updates

📞 **Queries?** WhatsApp our coordinator or email us — we respond within 24 hours.
📧 biomedical@shobhituniversity.ac.in | 📱 Join our WhatsApp group for instant updates